# Patrick Ming

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## Education

**Washington University in Saint. Louis** Aug 2025 - Exp. Dec 2028
*Computer Science and Mathematics* *St. Louis, Missouri*

- **Relevant Coursework: MATH 4502** - Topics in Applied Mathematics, Machine Learning Methods in Scientific Computing; **MATH 4301** - Linear Algebra; **MATH 4101** - Introduction to Analysis. **CSE 5403** - Algorithms for Nonlinear Optimization; **MATH 3300** - Matrix Algebra; **MATH 3010** - Foundations for Higher Mathematics; **MATH 2130** - Calculus III. **CSE 5100** - Deep Reinforcement Learning; **CSE 2407** - Data Structures and Algorithms; **CSE 2400** - Logic and Discrete Mathematics. **SDS 5010** - Probability; **SDS 3020** - Elementary to Intermediate Statistics and Data Analysis `(GPA 4.0/4.0 on Mentioned Coursework)`

### Honors

- 2024 United States National Chemistry Olympiad (USNCO) Honors List Award (**t150** nation)
- 2024 Mathworks Math Modeling (M3) Challenge Semi-Finalists (top 1%)

## Experience

**Huang's Intelligence (HINT) Lab (*Tentative*)** August 2026 - Present
*LLM Post-Training Researcher* *St. Louis, MO*

- Investigating distributional mismatch of On-Policy Distillation in small LLMs. Post-trained teacher on student prefixes with Supervised Fine-Tuning (SFT) to reduce distributional mismatch without losing capability.
- Quantified training examples (TEs) sweet spot to narrow distribution mismatch with Qwen3-1.7B student. On GMS8k, Qwen-4B: 105 TEs, Qwen3-8B: 105 or 785 TEs, Qwen3-14B: 523+ TEs. Results show no major loss of teacher capability with fine-tuning (average deviation is +0.03% performance).

**Qognitive Inc.** May 2026 - July 2026
*Deep Reinforcement Learning Researcher* *New York City, NY*

- Independently designed a hybrid policy paradigm ($\varepsilon$-Clipped-SAC) for Reinforcement Learning (RL) problems that outperforms SAC and PPO on 9/11 Gymnasium environments tested.
- Built an RL-guided Agentic AI Quantum Algorithmic Cooling controller under a U.S. Department of Energy–funded project with novel hybrid.

## Projects

**Autonomous Build Orchestrator (Claude Code Plugin)** July 2026 - August 2026
*Agentic Systems, LLM Orchestration, Automated Verification* *On Github*

- Autonomous phase-based development pipeline (Plan; Build; Verify; Clean; Commit) that orchestrates subagents (Orchestror; Verifier-Builder; Worker; Repo Scanner; Finalizer).
- Used to build: Paper-to-Audiobook pipeline for research papers; Chemistry-RL environment; Canvas Course Agent

**RL Environment for Mathematical Construction/Counterexample Tasks** Jun 2026 - Now
*Post-Training, Reinforcement Learning, LLM Evaluation* *On Github*

- A 314-task RL environment and benchmark for math construction/counterexample problems (LinAlg, Algebra, Analysis). 31 hand-designed verifiers, gated against reward hacking with 1912 wrong-object acceptances.

**WUSTL Research Matchmaker** Jan 2026 - April 2026
*AI Research Assistant + Faculty Research Matchmaker* *On Github*

- 3 hr process turned into 5 minutes. Multi-vector RAG model for streamlining research discovery. Tabulated 1,584 STEM faculty using custom WUSTL webscraper.

## Miscellaneous, Skills, and Interests

Research Diary/Blog
**Languages**: Python (NumPy, PyTorch, Pandas, Matplotlib, Flet, CustomTkinter), R, Java, LaTeX
**Technologies**: Excel, Overleaf, VS Code, Pycharm
**Research Interests:** Large Language Model Reasoning; Applied Machine Learning; On-Policy Distillation; Post-Training; Deep Reinforcement Learning, Deep Learning; Artificial Intelligence/Machine Learning.